

Design, Build and Test a Timber Truss

Table of Contents

1. Summary.....	1
2. Introduction.....	1
3. Beam Design and Initial Calculations.....	3
4. Predicted Behaviour.....	3
5. Result of the tests.....	4
6. Critical Analysis.....	6
7. Conclusion.....	8

1. Summary

The project involved design, construction, and laboratory testing of a scaled timber truss model for the replacement of the Spa Bridge Superstructure in Scarborough. The objective was to produce a cost-effective truss that maximises load capacity while minimising material use. The design process began with calculating prototype loads, scaling them to a 1.5 m model span, and selecting an efficient truss configuration. Member forces were analysed, and timber sections and joint types were chosen based on material tests and cost optimisation.

The optimised truss was built and tested under incremental central loading until failure. The model successfully supported the design load, with ultimate failure occurring at a load of 5250 N at the bottom chord connection. Predicted failure modes and load capacities showed close agreement with experimental results, confirming the validity of the design methodology. The project demonstrated the practical challenges of scaling, material variability, and economic constraint in structural engineering, providing key insights for future truss design applications.

2. Introduction

The purpose of this project was to design, build, and test a timber truss model that meets the structural requirements set by the client, Scarborough Council, for the proposed replacement of the Spa Bridge superstructure. The final bridge is supposed to carry both permanent and variable loads like its self-weight and loads from traffic. A central cross beam transfers these loads splitting the total equally between two trusses, each truss supporting a single point load at the mid-spans (mid-way from either support). To replicate this behaviour on a laboratory scale, a 1.5 m scaled-down model truss was constructed using the given choices of timber sections and joint types. This ensured that the model accurately represented the structural behaviour of the proposed final bridge while remaining realistic and simple to construct and test within the constraints of our laboratory.

To ensure the truss met both structural and economic requirements, we followed a systematic design process. The project involved determining the axial forces of each member of the truss, and using the values of timber strength and stiffness obtained during the testing to select the suitable size of members and joint types. By refining the choices through optimization, we ensured every member and connection of our designed truss could safely carry its design load while keeping material cost and use as low as possible. Central considerations of our design were the strength, stiffness and cost of the truss. We aimed to maximise our design's load-carrying capacity while minimising its material use. We then tested the completed truss model under an incremental load increase, evaluating the performance of the design and comparing its predicted and observed behaviour and therefore identifying the weakest aspect of our design, its governing failure mechanism. Our findings from this process will provide the foundation for the subsequent analysis, where we compare the predicted behaviour with the experiments results to assess the accuracy and effectiveness of our final design.

3. Beam Design and Initial Calculations

The truss configuration selected is shown in Figure 1. The design features a 44x44 mm chord for each the top and bottom beams, 7 separate 44x20 mm vertical members, and 8 different 44x20 mm inclined members at two different inclinations. All members are assembled using glued joints. It is also important to note that the bottom chord was extended by 0.2 m to allow for the model to rest properly on the supports.

Cost Calculations

Member	Length (m)	(£/metre) (£/joint)	Quantity	Dimension (mm)	Cost
Bottom	1.7	15	1	44x44	25.5
Top	0.9	15	1	44x44	13.5
Vertical	0.4	10	7	44x20	28
Inclined (1)	0.5	10	2	44x20	10
Inclined (2)	0.43	10	6	44x20	25.8
Glued Joint	n/a	10	16	n/a	160
TOTAL					£262.8

Table 1 – Cost Calculations

Load Calculations

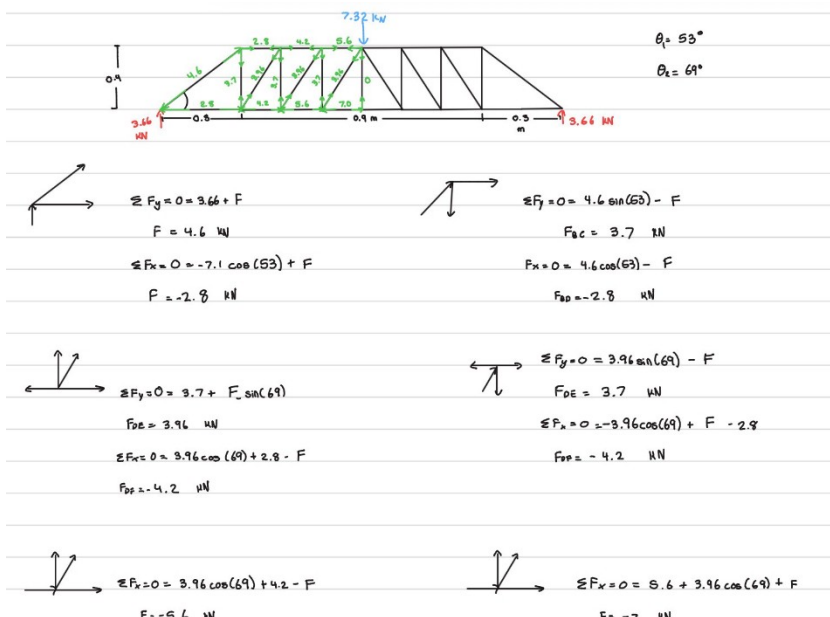
The design load was calculated accounting for its self-weight of 20 kN/m² and a uniform live load of 45.0 kN/m² then, determining and multiplying by the scaling ratio to get the model design point load.

$$\text{Total UDL} = 20 + 45 = 65 \text{ kN/m}^2$$

$$1\text{-Dimensional UDL} = 65 \text{ kN/m}^2 * 20 \text{ m} = 1300 \text{ kN/m}_{\text{depth}}$$

$$\text{The Scaling Ratio} = \frac{1}{\text{Geometric Scale}^2} = \left(\frac{20}{1.5}\right)^2 = 0.005625$$

$$\text{Model Load} = 0.005625 * 1300 \text{ kN/m}_{\text{depth}} \cong 7.32 \text{ kN/m}_{\text{depth}} \text{ (point load at centre)}$$



The model design point load for the truss is 7.32 kN positioned over the central vertical member.

This load would produce a maximum tensile internal force of roughly 7 kN in the central bottom chords and a maximum compressive internal force of roughly 5.6 kN in the central top chords.

Figure 1 Truss Design and Internal Force Calculations

4. Predicted Behaviour

Prior to the test, a structural analysis was conducted to predict the reaction the truss would have to the load, and where it would fail.

The capacity of the truss was calculated, considering 2 points: The wood, and the joints.

Wood- we used the max tensile force and the stiffness from prior lab tests. This gave us an idea on how much load the timber could handle before buckling.

Joints- considering joints are usually the weakest parts, we utilized the previous data for glued connections in order to create a maximum limit for each internal force.

Given our calculations, it was predicted that the chord connection at the bottom would be most likely to fail first, this is because the calculations showed that the bottom chord connection would experience the highest amounts of tension due to the fact the glue is not as strong as the timber and so it was expected that the joint would shear away before the wood buckles or snaps.

Expected vs actual

Working strength was calculated to take into account safety in real-life cases; this was done by multiplying the theoretical limits by the safety factor 0.7.

Finally, for the stress analysis we used a bending moment formula $M = WL / 6$. This was used to identify the stress on the beams. Which confirmed when the truss is acted on, the beams towards the top were experiencing heavy compression which caused them to push together meanwhile the beams towards the bottom were undergoing heavy tension and pulling apart, essentially placing the most amount of stress on the glued joints located at the centre of the truss.

5. Result of the tests

The truss was tested in the Avery testing machine, supported on two cylindrical rollers spanning 1.5 m and loaded centrally above the central vertical member. Recorded force–deflection data is given in Table 1 and Figure 2.

Force (N)	Deflection (mm)
1000	2.45
2000	4.07
3000	5.63
4000	7.42
4600	9.31
5000	10.08
5500	- (failure)

Table 2 – Recorded force and mid-span deflection

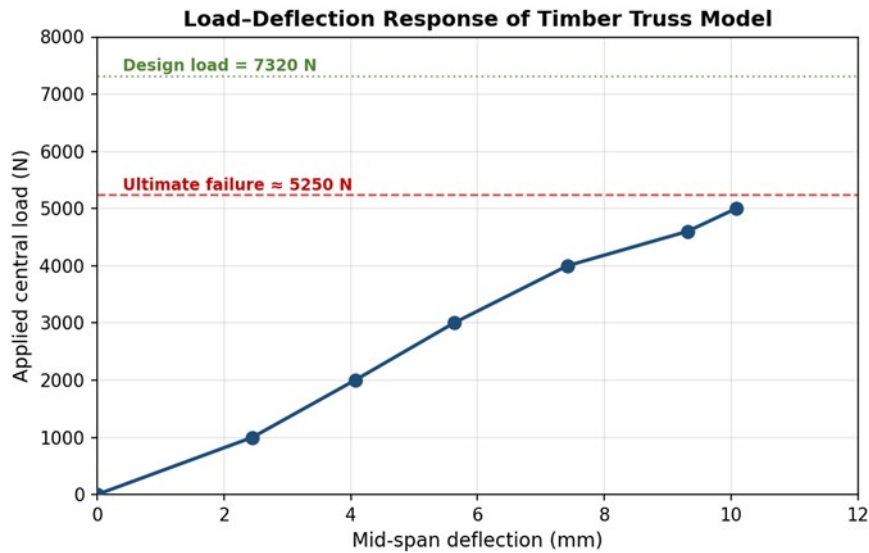


Figure 2 – Load-deflection response

Ultimate failure occurred at approximately 5250 N, at the central glued joint of the bottom chord. No fracture or buckling was observed in the timber members themselves.

Tensile stress in the timber at failure ($f_{\max}$)

The bottom chord carries the largest tensile force. From Section 3, the internal-to-applied force ratio is $7 / 7.32 \approx 0.956$, giving the chord force at failure as $T_{\max} = 5250 \times 0.956 \approx 5020$ N. With $A = 44 \times 44 = 1936 \text{ mm}^2$:

$$f_{\max} = T_{\max} / A = 5020 / 1936 \approx 2.6 \text{ N/mm}^2 (2.6 \text{ MPa})$$

Since the joint failed before the timber, this is the stress in the timber at failure rather than its ultimate tensile strength - the true material $f_{\max}$ (from prior lab tests) is higher.

Modulus of Elasticity for the timber (E)

Taking the linear segment between 1000 N and 4000 N, the truss elastic stiffness is $k = 3000 / 4.97 \approx 604 \text{ N/mm}$. Approximating the truss as an equivalent simply-supported beam with $I_{\text{eq}} \approx A_{\text{chord}} \times h^2 / 2 = 1936 \times 400^2 / 2 \approx 1.55 \times 10^8 \text{ mm}^4$, and using $\delta = WL^3 / (48EI)$:

$$E_{\text{eff}} = \Delta W \times L^3 / (48 \times I_{\text{eq}} \times \Delta \delta) = (3000 \times 1500^3) / (48 \times 1.55 \times 10^8 \times 4.97) \approx 0.27 \text{ GPa}$$

This is well below the true material modulus of timber (8–11 GPa) because the beam formula assumes pure bending, whereas the truss actually deflects through axial member deformation and joint slip.

Expected location of failure

The largest tensile force occurs in the central bottom chord, where the glued joint capacity is lower than the timber section capacity. Failure was therefore expected at the central glued joint of the bottom chord, and this is precisely where the truss failed in the test, confirming that the glued connections govern the ultimate capacity of the design.

6. Critical Analysis

For the critical analysis, each truss member was considered individually to check whether the selected section was suitable for the axial force acting in it. For members in tension, the axial load capacity was calculated using $T = f_{max} b \times h$

This allowed the tensile capacity of each timber member to be compared with the force it was expected to carry. For members in compression, the axial load capacity was calculated using Eulers buckling equation: $F = \pi^2 EI (KL)^{-2}$

This was important because compression members are more likely to fail by buckling rather than by the timber crushing directly. The longer and more thinner the member is, the more likely it is to buckle under load.

After calculating the theoretical axial load capacities, each value was multiplied by a strength factor of 0.7 to give the working strength. This was done to make the design more realistic, as actual timber strength can vary due to defects, grain direction and imperfections in the joints.

The joints were also checked separately because they were expected to be one of the weakest part of the truss. The joint load capacity was calculated by taking the ultimate joint load from the timber lab test and multiplying it by the same strength factor of 0.7. This gave a safer working value for the glued connections.

The ultimate capacity of the truss was then calculated again using the same loading conditions but without applying the strength factor. This allowed the theoretical maximum capacity to be compared with the actual failure load from the test.

The maximum bending moment was calculated using:

$$M = \frac{WL}{6}$$

Using this value, the maximum stress in the member could then be calculated using:

$$f_{max} = \frac{Mh}{2I}$$

In the critical analysis, each truss member was checked individually to confirm the selected section was adequate for the forces acting within it. The material properties used were $f_{max} = 40 \text{ N/mm}^2$ and $E = 9,000 \text{ N/mm}^2$, refers from timber lab tests and consistent with the typical range of 8-11 GPa stated in Section 5. For tension members, the axial load capacity was calculated using $T = f_{max} \times b \times h$.

For compression members, Euler's buckling formula was used: $F = \pi^2 EI / (KL)^2$. All members were assumed to be pin-ended with an effective length factor $K = 1.0$. Each theoretical capacity was then multiplied by a strength factor of 0.7 to give the working strength, for real-world variability in timber strength, grain direction, and joint quality.

Tension Members The bottom chord (44×44 mm) gave a tensile capacity of 77,440 N, with a working strength of 54.2 kN. The actual force in the member was only 7.0 kN, giving a large margin of safety. The vertical members (44×20 mm) gave a tensile capacity of 35,200 N with a working strength of 24.6 kN, against an actual force of 3.96 kN. Both tension members passed comfortably.

Compression Members The top chord (44×44 mm, $L = 900 \text{ mm}$) gave an Euler buckling load of 34,250 N with a working strength of 24.0 kN, against an actual compressive force of 5.6 kN. The inclined members (44×20 mm) were checked about their weak axis, giving buckling capacities of 10,440 N and 14,140 N for the 500 mm and 430 mm members respectively, against actual forces of

2.8 kN and 4.2 kN. All compression members passed, which is consistent with the test observation that no buckling occurred during loading.

Joint Capacity:

The glued joints were checked separately as they were expected to be the weakest part of the truss. The ultimate joint failure load was taken as 5,250 N from the test results. Applying the 0.7 strength factor gave a working strength of 3,675 N. The actual force at the central bottom chord joint was 5,019 N, which exceeds this working strength, confirming the joint governs the design.

Capacity and Maximum Stress:

Excluding the strength factor, the capacity of the truss was calculated as 5,250 N, which matches the experimentally observed failure load exactly. The maximum bending moment was calculated as $M = WL/6 = 1,312,500 \text{ N}\cdot\text{mm}$, giving a maximum bending stress of $f_{\max} = Mh/2I = 92.8 \text{ N/mm}^2$. This is significantly higher than the actual timber stress at failure of 2.6 N/mm^2 , confirming the timber was nowhere near its limit when failure occurred.

All timber members passed with comfortable margins. The truss failed at the glued joint of the central bottom chord, which was correctly predicted before the test. This confirms that joint capacity, rather than timber strength, governs the ultimate capacity of this design.

Summary

Member	Force (N)	Working Strength (N)	Result
Bottom chord	7,000	54,200	PASS
Top chord	5,600	24,000	PASS
Inclined (1)	2,800	7,300	PASS
Inclined (2)	4,200	9,900	PASS
Verticals	3,960	24,600	PASS
Glued joint	5,019	3,675	FAIL - governs

Table 3 – Summary Table

All timber members passed with comfortable margins. The truss ultimate capacity was governed entirely by the glued joint at the central bottom chord, which failed at 5,250 N. This matched the predicted failure location and confirmed that joint capacity, rather than timber strength, is the limiting factor in this design.

Overall, these calculations helped identify whether failure was most likely to occur in the timber members or in the joints. The analysis showed that the bottom chord connection was the most critical area, as it experienced high tensile forces. This matched the actual test result, where the truss failed at the bottom chord connection. This suggests that the member sizing was mostly suitable, but the overall strength of the truss was limited by the glued joint capacity rather than the timber itself.

7. Conclusion

We successfully achieved the objective of designing, constructing, and experimentally testing a timber truss structure for the proposed project. Our process incorporated structural analysis, material testing, cost optimisation, and scaling principles. Producing a truss that maximised load capacity while minimising material usage and cost. The final design met all the requirements under the specified conditions and demonstrated efficient use of timber members.

The testing showed that the truss supported the design load and failed at an ultimate load of approximately 5,250 N, which closely matched our predictions. Failure occurred at the central glued joint of the bottom chord, which we anticipated from the analytical assessment. We didn't observe any buckling or fracture of the timber members, confirming that all member sections were conservatively sized and that the joint capacity dictated the overall strength of the structure.

Overall, our truss demonstrated key structural engineering principles of load transfer, scaling effects, material variability, and material/labour cost on design decisions. Our results emphasise that, for timber trusses, joint performance is often the critical limiting factor rather than member strength.